

Complex Analysis – 4

Вычеты: интегралы и суммирование рядов

Домашнее задание

1 сентября 2026 года

Теоретический минимум. Laurent expansion and residues; the residue theorem; standard contours (semicircle, rectangle, sector, keyhole, indented contour); Jordan's lemma; summation of series through $\pi \cot \pi z$ and $\pi / \sin \pi z$; the residue at infinity and the identity $\sum_{\text{all}} \text{res} = 0$ for a rational function.

1. Compute the Fresnel integrals $\int_0^\infty \cos(x^2)dx = \int_0^\infty \sin(x^2)dx = \frac{1}{2}\sqrt{\frac{\pi}{2}}$ by integrating e^{iz^2} over the sector of angle $\frac{\pi}{4}$.
2. Compute $\int_0^\infty \left(\frac{\sin x}{x}\right)^2 dx$ and $\int_0^\infty \frac{x^{a-1}}{1+x^n} dx$ for $0 < a < n$.
3. Prove that $\sum_{n \geq 1} \frac{1}{n^4} = \frac{\pi^4}{90}$ by the cotangent method, and find the general formula for $\sum n^{-2k}$ in terms of Bernoulli numbers.
4. Compute $\int_{-\infty}^\infty e^{-x^2} \cos(bx) dx$ by integrating e^{-z^2} over a rectangle, and explain why no residue is involved.
5. (Proposed problems) Prove that $\int_0^\infty \frac{\cos x - \cos(tx)}{x} dx = \ln |t|$ for every $t \neq 0$.
6. Prove that the sum of all residues of a rational function, including the residue at infinity, is zero, and use this to evaluate $\int_{-\infty}^\infty \frac{x^2}{x^4+1} dx$ in two ways.
7. Compute $\int_0^\pi \ln(\sin \theta) d\theta = -\pi \ln 2$ by a contour argument, and compare with the real proof through the doubling formula.
8. Let f be meromorphic on $\mathbb{C}$ with finitely many poles, none of them integers, and $f(z) = O(|z|^{-2})$. Prove that $\sum_{n \in \mathbb{Z}} f(n) = -\sum_{\text{poles}} \text{res}(\pi \cot(\pi z) f(z))$, and apply it to $f(z) = \frac{1}{(z+a)^2}$.

9. Derive the formula $\frac{1}{2\pi i} \oint \frac{f'}{f} z^k dz = \sum a_j^k$ for the power sums of the zeros of a polynomial, and use it to express $\sum a_j^2$ through the coefficients.
10. Prove the Poisson summation formula $\sum_{n \in \mathbb{Z}} f(n) = \sum_{k \in \mathbb{Z}} \hat{f}(k)$ for a Schwartz function f , and check it on $f(x) = e^{-\pi t x^2}$, obtaining the theta transformation $\theta(1/t) = \sqrt{t} \theta(t)$.